How BARCS Finds Important Genes

A student guide with worked examples and a quiz

Prepared for Hyun-Hwan Jeong and Kyu-Won Lee

BARCS version 0.2.1 · Reading level: grades 7 to 9

About this guide. This guide explains how the BARCS computer program works. You need only arithmetic, fractions, percents, powers and square roots. The guide teaches logarithms when it uses them. Each section uses the same small set of numbers, so you can follow every step. The guide uses the writing rules of ASD-STE100 Simplified Technical English.

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1 The experiment: a CRISPR screen

Each cell in your body has about 20,000 genes. A gene is a set of instructions. Scientists want to know which genes a cell needs, for example to survive a cancer drug.

A **CRISPR screen** tests thousands of genes at the same time. It works in four steps:

1. Scientists make a **library** of short pieces of RNA, called **guides**. Each guide switches off one gene.
2. They put one guide into each cell. Now each cell has one gene switched off.
3. They grow the cells. Some cells grow under normal conditions (**control**). Other cells grow with a drug (**drug**).
4. A sequencing machine reads the guides in each group. The machine counts how many times it reads each guide. This number is a **read count**.

If cells need a gene to survive the drug, the cells that lose that gene die. Their guide becomes rare in the drug group. So the guide count goes down.

Remember

BARCS does not look at cells directly. It looks at read counts. A guide that becomes rare tells us that its gene is important.

Scientists repeat each group more than one time. Each repeat is a **replicate**. Each sample that the machine reads is a **library**. In this guide, we use four libraries: two control replicates and two drug replicates.

2 The data

The machine gives two numbers for each library:

- the read count of our guide, and
- the total read count of all guides in that library.

Table 1 shows the data for one guide. We use these numbers in all sections of this guide.

What the symbols mean

K	The read count of one guide in one library. Library 1 has $K = 180$.
N	The total read count of all guides in one library. Library 1 has $N = 1{,}000{,}000$.
i	The library number (1, 2, 3 or 4). We write K_i for the count in library i. So $K_2 = 240$.

Table 1: Read counts for one guide in four libraries.

Library	Group	Guide count K	Total count N
1	control	180	1,000,000
2	control	240	1,200,000
3	drug	120	1,000,000
4	drug	80	800,000

Remember

Always record N before you remove any guides from the table. If you remove guides first, the totals become wrong. Then all the later numbers become wrong too.

3 Step 1: Turn counts into proportions

Library 2 has more reads than library 1 because the machine read it more. So we cannot compare raw counts. We divide each count by its library total. The result is a **proportion**:

$$p = \frac{K}{N}.$$

Proportions are very small numbers. So we multiply by one million. The result is the count **per million** reads.

Worked example 1: proportions

Library 1: $p = 180 \div 1{,}000{,}000 = 0.000180$ = 180 per million
Library 2: $p = 240 \div 1{,}200{,}000 = 0.000200$ = 200 per million
Library 3: $p = 120 \div 1{,}000{,}000 = 0.000120$ = 120 per million
Library 4: $p = 80 \div 800{,}000 = 0.000100$ = 100 per million

The two control libraries give 180 and 200 per million. The two drug libraries give 120 and 100 per million. The guide is less common with the drug.

What the symbols mean

p The proportion of all reads that belong to the guide. It is always between 0 and 1.

4 Step 2: Compare the groups with a fold change

We want one number that tells how much the guide changes. First, we combine the replicates of each group. We add their counts and add their totals:

$$p_{\text{control}} = \frac{180 + 240}{1{,}000{,}000 + 1{,}200{,}000} = \frac{420}{2{,}200{,}000} = 0.000191 \text{ (191 per million)},$$

$$p_{\text{drug}} = \frac{120 + 80}{1{,}000{,}000 + 800{,}000} = \frac{200}{1{,}800{,}000} = 0.000111 \text{ (111 per million)}.$$

The **fold change** is the drug proportion divided by the control proportion:

$$\text{fold change} = \frac{p_{\text{drug}}}{p_{\text{control}}} = \frac{0.000111}{0.000191} = 0.58.$$

A fold change of 0.58 means that the guide keeps only 58% of its control level. It loses 42%.

Remember

- Fold change $= 1$: no change.
- Fold change < 1: the guide becomes rarer. Cells may need this gene.
- Fold change > 1: the guide becomes more common. Losing this gene may help the cells.

4.1 Logarithms make changes symmetric

A fold change of 2 means "double". A fold change of 0.5 means "half". These changes have the same size, but the numbers 2 and 0.5 do not look equal. A **logarithm** fixes this problem.

The **base-2 logarithm**, $\log_2$, tells how many times you double.

Fold change	0.25	0.5	1	2	4
$\log_2$(fold change)	-2	-1	0	1	2

Now "double" is $+1$ and "half" is -1. The sizes look equal.

Worked example 2: log fold change

Our fold change is 0.58. It is between 0.5 and 1. So $\log_2(0.58)$ is between -1 and 0.
A calculator gives $\log_2(0.58) = -0.78$.
The guide drops by 0.78 "halvings".

BARCS uses the **natural logarithm**, ln. It counts powers of the number $e = 2.718\ldots$ instead of powers of 2. The idea is the same. To change back from ln, you raise e to that power: if $\ln(x) = b$, then $x = e^b$. For our guide, $\ln(0.58) = -0.54$.

5 Step 3: Odds and log-odds

BARCS works with **odds** instead of proportions. The odds compare the reads that belong to the guide with the reads that do not:

$$\text{odds} = \frac{p}{1-p}.$$

The **log-odds** is the natural logarithm of the odds:

$$\text{log-odds} = \ln\left(\frac{p}{1-p}\right).$$

Worked example 3: odds and log-odds

Control: $p = 0.000191$. So odds $= 0.000191 \div 0.999809 = 0.000191$.
The odds and the proportion are almost equal. This is true for all small proportions, because $1 - p$ is almost 1.
Log-odds for control: $\ln(0.000191) = -8.56$.
Log-odds for drug: $\ln(0.000111) = -9.10$.
Difference: $-9.10 - (-8.56) = -0.54$.
The difference of the log-odds is -0.54. This is the same as ln(fold change) from Step 2.

Remember

For guides, a difference in log-odds is almost exactly the natural log of the fold change. So you can read BARCS results as log fold changes.

Why use log-odds at all? A proportion must stay between 0 and 1. A log-odds can be any number, positive or negative. This property lets BARCS add effects together safely, as Step 5 shows.

6 Step 4: Two kinds of noise

The two control libraries gave 180 and 200 per million. They are not equal. Replicates always differ a little. We call this difference **noise**. BARCS must know how big the noise is. Then it can tell a real change from a chance change.

Guide counts have two kinds of noise:

1. **Counting noise.** The machine reads a random sample of all the guides. Think of a bag with a million marbles. You take out a handful. You do not get exactly the same number of red marbles each time.
2. **Biological noise.** Each replicate is a different group of living cells. The cells grow a little differently each time.

BARCS measures the size of the noise with the **variance**. The variance is the average squared distance of a value from its expected value. The **standard deviation** is the square root of the variance. It is the typical distance in the same units as the counts.

BARCS uses this formula for the variance of a guide count:

$$\text{variance} = \underbrace{N\,\mu\,(1-\mu)}_{\text{counting noise}} \times \underbrace{\big(1+(N-1)\,\rho\big)}_{\text{extra biological noise}}.$$

What the symbols mean

μ	"mu". The true proportion of the guide that we expect. For the control group, our best value is $\mu = 0.000191$.
N	The library total, as before.
ρ	"rho". A number that measures the biological noise. If $\rho = 0$, only counting noise exists. A bigger ρ means more biological noise.

Worked example 4: variance

Use N = 1,000,000 and μ = 0.000191. For our guide, BARCS estimates ρ = 0.000000386.
Counting noise: $N\mu(1-\mu) = 1{,}000{,}000 \times 0.000191 \times 0.999809 = 191$.
Extra factor: $1 + (N-1)\rho = 1 + 999{,}999 \times 0.000000386 = 1.39$.
Total variance: $191 \times 1.39 = 265$.
Standard deviation: $\sqrt{265} = 16.3$ reads.
So a count of about 191 normally moves up or down by about 16. Biological noise adds 39% to the counting noise.

Remember

Look at the extra factor $1 + (N-1)\rho$. It contains N. When the machine reads more, N grows, and the factor grows too. So more reading does not remove biological noise. Only more replicates help with biological noise.

7 Step 5: The model

A **model** is a rule that predicts a value. BARCS predicts the log-odds of a guide in each library. For our two groups, the rule is:

$$\text{log-odds} = \beta_0 + \beta_1 \times x.$$

What the symbols mean

x	A label for the group. $x = 0$ for control. $x = 1$ for drug.
β_0	"beta zero". The log-odds of the guide in the control group. It is the starting level.
β_1	"beta one". The change in log-odds when you go from control to drug. This is the **effect**. It is the most important number.

This rule is like the equation of a line, $y = b + mx$. The value β_0 is the starting point. The value β_1 is the slope.

BARCS finds the values of β_0 and β_1 that fit the four libraries best. It also uses the noise from Step 4. Libraries with less noise get more weight.

Worked example 5: the model for our guide

BARCS gives these values:
$\beta_0 = -8.565$ (control log-odds)
$\beta_1 = -0.542$ (effect of the drug)
Check the control level. Change -8.565 back to a proportion. $e^{-8.565} = 0.000191$, which is 191 per million. This agrees with Step 2.
Read the effect. Change β_1 back to a fold change: $e^{-0.542} = 0.58$. The drug keeps 58% of the guide. This agrees with Step 2.

BARCS can use more than one label. For example, it can include the replicate number, the day, or the drug dose. It adds one β for each label. This is the main advantage of BARCS. Older methods compare only two groups.

8 Step 6: Is the effect real?

The effect is $\beta_1 = -0.542$. But replicates are noisy. Maybe the true effect is 0 and the noise made the drop. BARCS tests this question in three parts.

8.1 The standard error

The **standard error** (SE) tells how much the estimate of β_1 can change because of noise. A small SE means a precise estimate. For our guide, BARCS gives SE = 0.101.

8.2 The t value

The **t value** compares the effect with its standard error:

$$t = \frac{\beta_1}{\text{SE}}.$$

A t value far from 0 means that the effect is large compared with the noise.

8.3 Degrees of freedom

The **degrees of freedom** (df) tell how much information we have about the noise:

$$\text{df} = (\text{number of libraries}) - (\text{number of } \beta \text{ values}).$$

Fewer degrees of freedom mean less information. Then BARCS needs a bigger t value before it accepts an effect as real.

8.4 The p-value

The **p-value** answers this question: "If the true effect is 0, how often does noise alone give a t value this far from 0?" A small p-value means that noise alone seldom gives such a result. Scientists often use 0.05 as a limit.

Worked example 6: testing the effect

1. $t = -0.542 \div 0.101 = -5.38$.
2. We have 4 libraries and 2 β values. So $\text{df} = 4 - 2 = 2$.
3. With 2 degrees of freedom, a t value must be farther from 0 than 4.30 to give $p < 0.05$. Our t is -5.38. It is farther from 0 than 4.30.
4. A computer gives the exact p-value: $p = 0.033$.

The p-value is less than 0.05. So the drop is probably not only noise.

Remember

The degrees of freedom come from the number of libraries, not from the number of reads. A library with 100 million reads still counts as one library.

9 Step 7: Borrow information from other guides

With only 2 degrees of freedom, the noise estimate of one guide is not reliable. But a screen has thousands of guides. Most guides have a similar amount of noise. So BARCS combines two estimates:

- the noise of this guide, and
- the typical noise of guides with similar counts.

This method is **moderation**. BARCS takes a weighted average. Each estimate gets a number of "votes". The guide's own estimate gets its degrees of freedom as votes. The typical estimate gets the **prior degrees of freedom**, d_0. BARCS calculates d_0 from all the guides.

$$\text{moderated noise} = \frac{\text{df} \times \text{own noise} + d_0 \times \text{typical noise}}{\text{df} + d_0}.$$

Worked example 7: moderation

A guide has 3 degrees of freedom. Its own noise value is 4.0. The typical noise value is 1.5, with $d_0 = 12$ votes.

$$\text{moderated noise} = \frac{3 \times 4.0 + 12 \times 1.5}{3 + 12} = \frac{12 + 18}{15} = \frac{30}{15} = 2.0.$$

The guide's noise moves from 4.0 toward 1.5. It stops at 2.0.

BARCS uses moderation when it has 50 or more usable guides. Moderation also adds the d_0 votes to the degrees of freedom. So the test in Step 6 becomes more powerful.

10 Step 8: Check the results with control guides

A good screen contains **non-targeting control guides**. These guides switch off no gene. So their true effect is 0.

If the noise model is correct, about 5% of control guides get $p < 0.05$ by chance. If many more control guides get $p < 0.05$, the noise model is too small. Then BARCS makes all the tests wider.

Worked example 8: calibration with control guides

A screen has 100 control guides. We expect about 5 of them to get $p < 0.05$. But 15 of them get $p < 0.05$. The noise is too small.
BARCS finds a **scale factor** that makes the control guides behave correctly. Suppose that the scale factor is 1.26. BARCS divides every t value by 1.26.
A gene with $t = 3.5$ now gets $t = 3.5 \div 1.26 = 2.78$. Its p-value becomes larger. This change protects you from false results.

The scale factor is never less than 1. So calibration can only make the results more careful.

11 Step 9: From guides to genes

Each gene usually has several guides, often 3 to 6. Each guide gives its own result. BARCS combines them into one result for the gene. It uses the **Stouffer method** in three parts.

1. Change each guide's p-value into a **z score**. A z score tells how many standard deviations a result is from 0. The sign of the z score follows the sign of the effect.
2. Add the z scores.
3. Divide the sum by the square root of the number of guides, m.

$$Z_{\text{gene}} = \frac{z_1 + z_2 + \cdots + z_m}{\sqrt{m}}.$$

Worked example 9: combining three guides

A gene has $m = 3$ guides. Their p-values are 0.046, 0.134 and 0.012. Their z scores are 2.0, 1.5 and 2.5.

$$Z_{\text{gene}} = \frac{2.0 + 1.5 + 2.5}{\sqrt{3}} = \frac{6.0}{1.732} = 3.46.$$

This Z gives $p = 0.00053$ for the gene. No single guide is that strong. Together, the three guides give strong evidence.

11.1 When guides share noise

Sometimes the guides of one gene share noise. Then they agree more than they should by chance. BARCS measures this sharing with the **guide correlation**, r. It makes the bottom of the fraction bigger:

$$Z_{\text{gene}} = \frac{z_1 + z_2 + \cdots + z_m}{\sqrt{m + m(m-1)\,r}}.$$

Worked example 10: shared noise

Use the same three guides and $r = 0.1$. The bottom becomes $\sqrt{3 + 3 \times 2 \times 0.1} = \sqrt{3.6} = 1.897$. So $Z_{\text{gene}} = 6.0 \div 1.897 = 3.16$, and $p = 0.0016$. The evidence is still strong, but less strong than before.

12 Step 10: Many genes and false discoveries

A screen tests about 20,000 genes. Suppose that no gene has an effect. If we use the limit $p < 0.05$, about 5% of the genes still pass by chance. That is about 1,000 false results.

So BARCS uses the **false discovery rate** (FDR). The FDR is the expected fraction of false results in your list of hits. BARCS calculates it with the **Benjamini–Hochberg method**:

1. Sort the genes from the smallest p-value to the largest.

2. Give each gene a rank: 1, 2, 3, and so on.
3. For each gene, calculate $p \times$ (number of genes) $\div$ rank.
4. Go from the bottom of the list to the top. If a value is larger than the value below it, replace it with the value below it.

Worked example 11: false discovery rate for five genes

Gene	Rank	p	$p \times 5 \div$ rank	FDR (after part 4)
A	1	0.001	$0.001 \times 5 \div 1 = 0.005$	0.005
B	2	0.008	$0.008 \times 5 \div 2 = 0.020$	0.020
C	3	0.039	$0.039 \times 5 \div 3 = 0.065$	0.051 (from D)
D	4	0.041	$0.041 \times 5 \div 4 = 0.051$	0.051
E	5	0.600	$0.600 \times 5 \div 5 = 0.600$	0.600

With the limit FDR < 0.05, genes A and B are hits. With the limit FDR < 0.10, genes A, B, C and D are hits.

13 Two special cases

13.1 A guide with zero reads

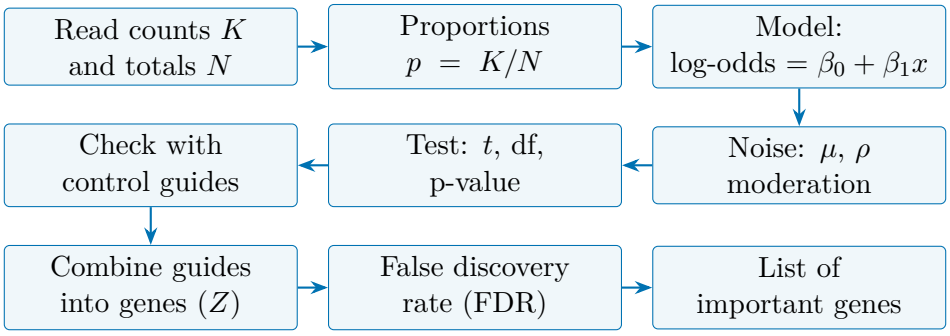
Sometimes a guide has 0 reads in all control libraries. Then the log-odds of the control group is $\ln(0)$. This value does not exist, because no power of e equals 0. Older versions of BARCS could not fit such a guide. So they removed it.

BARCS version 0.2.1 adds a very small number to the counts first. This number is less than one read. The method is the **Firth correction**. Now the guide gets a value, and BARCS keeps it.

13.2 A very large change

Sometimes a guide grows by hundreds of times. The usual test from Step 6 can then fail. The standard error grows faster than the effect, so the t value becomes small. BARCS version 0.2.1 has a second test, the **likelihood-ratio test**. It compares how well the model fits with the effect and without the effect. The option `test = "auto"` uses this test only for guides with very large changes.

14 Summary: the full path



All symbols in one table

Symbol	Meaning	Value in our example
K	read count of the guide in one library	180, 240, 120, 80
N	total read count of one library	1.0, 1.2, 1.0, 0.8 million
p	proportion, K/N	191 and 111 per million
fold change	$p_{\text{drug}} \div p_{\text{control}}$	0.58
$\log_2$	number of doublings	$\log_2(0.58) = -0.78$
ln	natural logarithm (powers of $e = 2.718$)	$\ln(0.58) = -0.54$
log-odds	$\ln(p/(1-p))$	-8.56 (control)
μ	expected proportion	0.000191 (control)
ρ	biological noise	0.000000386
x	group label (0 = control, 1 = drug)	0 or 1
β_0	control log-odds in the model	-8.565
β_1	effect of the drug	-0.542
SE	standard error of the effect	0.101
t	effect divided by SE	-5.38
df	libraries minus β values	$4 - 2 = 2$
p-value	chance of this result if the effect is 0	0.033
d_0	prior degrees of freedom (votes)	12 (Example 7)
z, Z	z score of a guide, of a gene	3.46 (Example 9)
m	number of guides for a gene	3
r	guide correlation (shared noise)	0.1 (Example 10)
FDR	expected fraction of false hits	0.005 for gene A

Note: the letter p has two uses in this guide. Alone, p is a proportion. In the word "p-value", it is a probability. The text always tells you which one it is.

15 Quiz

Choose the best answer. The answers and calculations are in the next section.

Q1. What does one guide do in a CRISPR screen?

(A) It counts the cells.

(B) It adds a drug to the cells.

(C) It reads the DNA.

(D) It switches off one gene.

Q2. A library has $K = 300$ and $N = 1{,}500{,}000$. What is the count per million?

(A) 200

(B) 300

(C) 450

(D) 500

Q3. Why do we divide each count by its library total N?

(A) To make the numbers bigger.

(B) Because the machine reads some libraries more than others.

(C) To remove the biological noise.

(D) Because BARCS cannot use whole numbers.

Q4. A guide has 400 per million in control and 100 per million with a drug. What is the fold change?

(A) 4

(B) 0.4

(C) 0.25

(D) 300

Q5. What is $\log_2(8)$?

(A) 2

(B) 3

(C) 4

(D) 8

Q6. A gene has $\log_2(\text{fold change}) = -1$. What happened to its guide?

(A) It doubled.

(B) It did not change.

(C) It dropped to half.

(D) It dropped to zero.

Q7. For a very small proportion p, which statement is true?

(A) The odds are almost equal to p.

(B) The odds are almost equal to 1.

(C) The odds are twice p.

(D) The odds are always 0.

Q8. In the variance formula, what does $\rho = 0$ mean?

(A) There is no noise at all.

(B) There is only counting noise.

(C) There is only biological noise.

(D) The guide has zero reads.

Q9. Use $N\mu(1-\mu)$ with $N = 100$ and $\mu = 0.5$. What is the counting variance?

(A) 25

(B) 50

(C) 0.25

(D) 100

Q10. The machine reads each library ten times more deeply. What happens to the biological noise?

(A) It disappears.

(B) It becomes ten times smaller.

(C) It does not go away. Only more replicates help.

(D) It becomes zero for control guides.

Q11. In the model log-odds $= \beta_0 + \beta_1 x$, which value is the effect of the drug?

(A) x

(B) β_0

(C) β_1

(D) the log-odds

Q12. BARCS gives $\beta_1 = 0.69$ for a guide. Use $e^{0.69} = 2.0$. What does this mean?

(A) The guide halves with the drug.

(B) The guide does not change.

(C) The guide disappears.

(D) The guide doubles with the drug.

Q13. An experiment has 6 libraries. The model has 3 β values. How many degrees of freedom does it have?

(A) 3
(B) 2
(C) 6
(D) 9

Q14. An effect is $\beta_1 = -0.8$ with SE $= 0.2$. What is the t value?

(A) -0.16
(B) -1.0
(C) -0.6
(D) -4.0

Q15. A guide has df $= 2$ and its own noise 5.0. The typical noise is 2.0 with $d_0 = 8$. What is the moderated noise?

(A) 2.0
(B) 3.5
(C) 5.0
(D) 2.6

Q16. A screen has 200 control guides. About how many should get $p < 0.05$ by chance?

(A) 0
(B) 5
(C) 10
(D) 50

Q17. A gene has 4 guides, each with $z = 1.0$. What is Z_{gene} (no shared noise)?

(A) 1.0
(B) 2.0
(C) 4.0
(D) 0.5

Q18. Three genes have p-values 0.01, 0.02 and 0.30. What is the Benjamini–Hochberg value $p \times 3 \div$ rank for the second gene?

(A) 0.02
(B) 0.06
(C) 0.30
(D) 0.03

Q19. Why does BARCS 0.2.1 add a very small number to the counts of some guides?

(A) Because ln(0) does not exist when a group has zero reads.

(B) To make all guides look important.

(C) To increase the library totals.

(D) To remove control guides.

Q20. Which action gives BARCS more information about biological noise?

(A) Read each library more deeply.

(B) Use fewer guides.

(C) Add more replicate libraries.

(D) Use a larger p-value limit.

16 Answers and calculations

Q1. D. Each guide switches off one gene (Section 1).

Q2. A. $300 \div 1{,}500{,}000 = 0.0002$. Then $0.0002 \times 1{,}000{,}000 = 200$ per million.

Q3. B. Libraries get different numbers of reads. Division by N makes them comparable (Step 1).

Q4. C. $100 \div 400 = 0.25$. The guide keeps one quarter of its control level.

Q5. B. $2 \times 2 \times 2 = 8$. You double three times, so $\log_2(8) = 3$.

Q6. C. -1 means one halving (Step 2).

Q7. A. $1 - p$ is almost 1, so $p/(1-p)$ is almost p (Step 3).

Q8. B. The extra factor becomes $1 + (N-1) \times 0 = 1$. Only counting noise stays (Step 4).

Q9. A. $100 \times 0.5 \times (1 - 0.5) = 100 \times 0.5 \times 0.5 = 25$.

Q10. C. The factor $1 + (N-1)\rho$ grows with N. Deeper reading does not remove biological noise (Step 4).

Q11. C. β_1 is the change from control ($x = 0$) to drug ($x = 1$) (Step 5).

Q12. D. The fold change is $e^{0.69} = 2.0$. The guide doubles.

Q13. A. df $= 6 - 3 = 3$ (Step 6).

Q14. D. $t = -0.8 \div 0.2 = -4.0$.

Q15. D. $(2 \times 5.0 + 8 \times 2.0) \div (2 + 8) = (10 + 16) \div 10 = 26 \div 10 = 2.6$ (Step 7).

Q16. C. 5% of $200 = 0.05 \times 200 = 10$ (Step 8).

Q17. B. $(1.0 + 1.0 + 1.0 + 1.0) \div \sqrt{4} = 4.0 \div 2 = 2.0$ (Step 9).

Q18. D. The second gene has rank 2. So $0.02 \times 3 \div 2 = 0.03$ (Step 10).

Q19. A. With zero reads, the log-odds is $\ln(0)$, which does not exist. The Firth correction fixes this problem.

Q20. C. Degrees of freedom come from libraries. Only more replicates add information about biological noise (Steps 4 and 6).